

Chapter: Introduction

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Exercise 1.7

Pattern Recognition and Machine Learning - Christopher M. Bishop

Solution

In this exercise, we prove the normalization condition (1.48) for the univariate Gaussian distribution:

$$\int_{-\infty}^{\infty} N(x|\mu, \sigma^2) dx = 1$$

where $N(x|\mu, \sigma^2)$ is the probability density function of a univariate Gaussian distribution with mean μ and variance σ^2 :

$$N(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

To do this consider, the integral:

$$I = \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}x^2\right) dx$$

which we can evaluate by first writing its square in the form:

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}x^2 - \frac{1}{2\sigma^2}y^2\right) dx dy$$

Now make the transformation from cartesian coordinates (x, y) to polar coordinates (r, θ) and then substitute $u = r^2$ to obtain:

Let's show that, by performing the integrals over θ and u , and then taking the square root of both sides, we obtain

$$I = 2\pi\sigma^{2^{1/2}}$$

Let's start by manipulating the expression for I^2

$$\begin{aligned} I^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}x^2 - \frac{1}{2\sigma^2}y^2\right) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2\sigma^2}(x^2 + y^2)\right) dx dy \end{aligned}$$

by performing the change of variables from cartesian coordinates (x, y) to polar coordinates (r, θ) :

$$I^2 = \int_0^{2\pi} \int_0^\infty \exp\left(-\frac{1}{2\sigma^2}r^2\right) r dr d\theta \quad \text{since } x^2 + y^2 = r^2 \text{ and } dx dy = r dr d\theta$$

Next, we perform the substitution $u = r^2$, which implies that $du = 2r dr$ or equivalently $r dr = \frac{1}{2} du$:

$$\begin{aligned} I^2 &= \int_0^{2\pi} \int_0^\infty \exp\left(-\frac{1}{2\sigma^2}u\right) \frac{1}{2} du d\theta \\ &= \frac{1}{2} \int_0^{2\pi} d\theta \int_0^\infty \exp\left(-\frac{1}{2\sigma^2}u\right) du \\ &= \pi \int_0^\infty \exp\left(-\frac{1}{2\sigma^2}u\right) du \quad \text{since } \int_0^{2\pi} d\theta = 2\pi \text{ and } \frac{1}{2} \cdot 2\pi = \pi \end{aligned}$$

Now we can evaluate the remaining integral over u :

$$\begin{aligned} \int_0^\infty \exp\left(-\frac{1}{2\sigma^2}u\right) du &= \left[-2\sigma^2 \exp\left(-\frac{1}{2\sigma^2}u\right) \right]_0^\infty \\ &= 2\sigma^2 \end{aligned}$$

Substituting this result back into the expression for I^2 , we have:

$$\begin{aligned} I^2 &= \pi \cdot 2\sigma^2 \\ &= 2\pi\sigma^2 \end{aligned}$$

Finally, by taking the square root of both sides, we obtain:

$$\begin{aligned} I &= \sqrt{2\pi\sigma^2} \\ &= 2\pi\sigma^{2^{1/2}} \end{aligned}$$

Finally by using this result, we show that the normalization condition holds:

$$\begin{aligned} \int_{-\infty}^\infty N(x|\mu, \sigma^2) dx &= \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^\infty \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^\infty \exp\left(-\frac{1}{2\sigma^2}z^2\right) dz \quad \text{by performing the change of variable } x - \mu \rightarrow z \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^\infty \exp\left(-\frac{1}{2\sigma^2}x^2\right) dx \quad \text{since } dz = dx \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} I \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} 2\pi\sigma^{2^{1/2}} \\ &= 1 \end{aligned}$$

Personal note: During the process of solving this exercise, I learned how to use the change of variables in multiple integrals (because of the integral operator properties) and how to evaluate the Gaussian integral by using polar coordinates and substitution.